Vaxcel®

OMEPRAZOLE 40MG INJECTION

MAL11040013AZ

COMPOSITION:

1 vial containing dry substance & 1 vial containing solvent. The dry substance for injection contains: Omeprazole Sodium equivalent to Omeprazole 40mg.

PRESENTATION:

A vial of white or almost white, hygroscopic powder packed with clear solvent that provides a clear colourless to clear light yellow solution after reconstitution.

INDICATIONS:

Duodenal ulcer, gastric ulcer, reflux oesophagitis and Zollinger-Ellison Syndrome.

PHARMACODYNAMICS:

Omeprazole an acid-inhibiting agents – proton pump inhibitors. Omeprazole is a substituted benzimidazole. Omeprazole is a racemate of two active enantiomers. The secretion of hydrochloric acid in the stomach is inhibited by omeprazole through its specific effect on the proton pump in the parietal cells. The effect on acid secretion is reversible. Omeprazole is a weak base, which is concentrated and converted into active form in the acidic environment in the parietal cell, where it inhibits H^+, K^+ ATpase, i.e. the final step in the production of the gastric acid. The inhibition is dose-dependent, and affects both basal and stimulated acid secretion, irrespective of the type of stimulation. Omeprazole does not affect cholinergic or histaminergic receptors. Like treatment with H_2-receptor blockers, treatment with omeprazole results in reduced acidity in the stomach and thus an increase in gastrin in proportion to the reduction in acidity. The gastrin increase is reversible. During long-term treatment the frequency of glandular cysts in the stomach may increase. These changes are physiological and consequence of the inhibition of acid secretion. They are benign and reversible. Decreased gastric acidity with proton pump inhibitors or other acid-inhibiting agents, increased the amount of bacteria normally present in the gastrointestinal tract, such treatment may lead to a slightly increased risk of gastrointestinal infections such as *Salmonella* and *Campylobacter*. No pharmacodynamic effects of clinical significance other than those due to the effect of omeprazole on acid secretion have been found. The effect on acid secretion is directly correlated to the area under the plasma concentration curve (AUC), but not to the actual plasma concentration of omeprazole. Intravenous administration of Vaxcel Omeprazole 40mg Injection results in an immediate reduction of hydrochloric acid secretion. A single dose of 40mg intravenously has approximately the same effect on the acidity of the gastric juice for 24 hours as a single oral dose of 80mg or repeated oral administration of 20mg once daily. During treatment with antisecretory medicinal products, serum gastrin increases in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumours. Available published evidence suggests that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

PHARMACOKINETICS:

Absorption: Binding to plasma proteins is approximately 95% and the volume of distribution is 0.3L/kg.

Metabolism: Omeprazole is metabolized completely, mainly in the liver. Mainly the enzymes CYP2C19 and CYP3A4 catalyse the metabolism. Identified metabolites are the sulphone, the sulphide and hydroxyl-omeprazole, which have no significant effect on the acid secretion. Total plasma clearance is 0.3 – 0.6 L/min.

Elimination: The half-life in plasma in the elimination phase is approximately 40 minutes (30 – 90 minutes) after multiple doses. Approximately 80% of the metabolites are excreted via the urine and the remainder in the feaces.

Patients Factors: Clearance is greatly reduced in patients with impaired liver function.

DOSAGE AND ADMINISTRATION:

Duodenal Ulcer, Gastric Ulcer and Reflux Oesophagitis:

patients who cannot be given oral medication can be treated parenterally with 40mg once daily. The usual treatment period before transfer to oral treatment is 2 - 3 days.

In Zollinger-Ellison syndrome the dose should be adjusted accordingly to individual. Higher doses and / or several doses daily may be required.

Special Populations:

Impaired Renal Function

A dose adjustment is not necessary for patients with impaired renal function.

Impaired Liver Function

In patients with impaired liver function clearance is greatly reduced.

Elderly Patients

Dose adjustment is not necessary in elderly patients.

Children

There is only limited experience of treatment in children.

ROUTE OF ADMINISTRATIONS:

Intravenous treatment can be given as injection, whereby the solution for injection must be given slowly over a period of at least 2 1/2 minutes and with a rate of maximum 4ml per minute. The solution for infusion is to be administered in an intravenous infusion for 20-30 minutes.

Instructions for Use and Handling:

Omeprazole Injection:

Solution for injection is obtained by dissolving the sterile dry substance with 10ml of solvent as shown in Table 1. The reconstituted solution for injection is stable up to 12 hours at room temperature (25^{o}C ± 5^{o}C).

Table 1: Preparation of Solution of Omeprazole Injection

No.	Diluent	Amount of diluent to be added in vial (ml)	Concentration (mg/ml)
1	Citrate Buffer	10ml	4mg/mL
2	Water for Injection	10ml	4mg/mL

Preparation:

1. With a syringe draw 10ml of the solvent.
2. Add approximately 5ml of the solvent to the vial with omeprazole dry substance.
3. Withdraw as much air as possible from the vial back into the syringe. This will make it easier to add the remaining solvent.
4. Add the remaining solvent into the vial, make sure the syringe is empty.
5. Rotate and shake the vial to ensure all the omeprazole has dissolved.

Omeprazole Infusion:

Solution for infusion is obtained by dissolving the sterile dry substance with one of the following diluents:

5% w/v dextrose solution for injection, 0.9% w/v sodium chloride solution for injection and 5% w/v mannitol, using the diluents volumes shown in Table 2.

Table 2: Preparation of Solution of Omeprazole Infusion

No.	Diluent	Amount of diluent to be added (ml)	Concentration (mg/ml)
1	5% w/v Dextrose solution for Injection	5ml and then immediately diluted to 100ml	0.4mg/mL
2	0.9% w/v Sodium Chloride solution for Injection	5ml and then immediately diluted to 100ml	0.4 mg/mL
3	5% w/v Mannitol	10ml water for injection and then immediately diluted to 100ml	0.4 mg/mL

Preparation:

1. With a syringe draw 5ml of infusion solution from the 100ml infusion bottle or bag.
2. Add this volume to the vial with the omeprazole powder, mix thoroughly making sure all omeprazole is dissolved.
3. Draw the omeprazole solution back into the syringe.
4. Transfer the solution into the infusion bag or bottle.
5. Repeat steps 1-4 to make sure all omeprazole is transferred from the vial into the infusion bag or bottle.

Compatibility and Stability:

Vaxcel Omeprazole-40mg Injection is compatible at concentrations between 0.4 and 4mg/ml with one of the following diluents: Citrate buffer, water for injection, 5% w/v dextrose solution for injection, 0.9% w/v sodium chloride solution for injection and 5% w/v mannitol. Vaxcel Omeprazole-40mg Injection reconstitution compatibility and stability information is summarized in Table 3.

Table 3: Vaxcel Omeprazole-40mg Injection Reconstitution Stability

No.	Concentration (mg/ml)	Diluent	Stability time at room temperature (25^{o}C ± 5^{o}C)
1	4mg/mL	Citrate Buffer	12 hours
2	4mg/mL	Water for Injection	12 hours
3	0.4mg/mL	5% w/v Dextrose solution for Injection	12 hours
4	0.4mg/mL	0.9% w/v Sodium Chloride solution for Injection	24 hours
5	0.4mg/mL	5% w/v Mannitol	24 hours

CONTRAINDICATIONS:

Known hypersensitivity to omeprazole. Omeprazole like other PPIs should not be administered with Atazanavir.

SIDE EFFECTS:

The most common symptoms that have been reported in clinical trials with Vaxcel Omeprazole 40mg Injection have been gastrointestinal, such as diarrhea, nausea and constipation, and also headache, each one in 1-3% of cases.

Common:

General: Headache

Gastrointestinal: Diarrhoea, nausea / vomiting, constipation, abdominal pain, flatulence, gland polyps (benign).

Less Common:

General: Fatigue

Skin: Rash, pruritus, urticaria

Liver: Changes in liver function tests

Neurological: Paraesthesia, dizziness, drowsiness

Psychiatric: Sleep disturbance

Musculoskeletal: Fracture of the hip, wrist or spine

Rare:

General: Increased sweating, peripheral oedema, hyponatraemia. Hypersensitivity reactions such as angiooedema, fever and anaphylactic shock.

Blood: Leukopenia, thrombocytopenia, agranulocytosis, pancytopenia

Endocrine: Gynaecomastia

Gastrointestinal: Dry mouth, taste disturbances, stomatitis, candidiasis

Skin: Hair loss, photosensitivity, erythema multiforme

Liver: Encephelopathy in patients with severe hepatic disease, hepatitis with or without jaundice, liver failure

Respiratory tract: Bronchospams

Muscosceletal: Arthralgia, muscular weakness and myalgia.

Psychiatric: Reversible confusion, agitation, depression, aggression and hallucinations especially in severely ill patients.

Isolated cases of Steven-Johnson syndrome and toxic epidermal necrolysis have been reported, but a relationship with omeprazole could not be established. In severe ill patients, in isolated cases irreversible visual disturbances have been reported in connection with treatment with high doses of omeprazole given as intravenous injection.
However, no causal relationship with omeprazole could be established.

SYMPTOMS AND TREATMENTS FOR OVERDOSAGE:

Intravenous doses up to 270mg in a day and up to 650mg over a three-day period have been studied in clinical trials without any doses-related adverse reactions.
Symptoms: Dizziness, apathy, headache, tachycardia, nausea, vomiting, flatulence, diarrhea. See also section Side effects.

WARNING AND PRECAUTION:

Suspected ulcer disease must be verified objectively at an early stage by means of X-ray or endoscopy in order to avoid inadequate treatment.
When gastric ulcer is present or suspected or in the presence of any of the following alarm symptoms: significant unintentional weight loss, recurrent vomiting, dysphagia, haematemesis or melena, malignancy should be excluded as treatment may alleviate symptoms and delay diagnosis.
Effects on ability to drive and use machines: Omeprazole is not likely to affect the ability to drive or use machines.

Pregnancy:
Well-conducted epidemiological studies indicate no adverse effects of omeprazole on pregnancy or on the health of the fetus / newborn child. Omeprazole can be used during pregnancy.

Lactation:
Omeprazole is excreted in breast milk. Influence, if any, on the child is unknown.

Regular Surveillance:
Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Subacute Cutaneous Lupus Erythematosus (SCLE):
Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Vaxcel Omeprazole 40mg Injection. SCLE after previous treatment with a proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Hypomagnesaemia:
Severe hypomagnesaemia has been reported in patients treated with PPI like Vaxcel Omeprazole 40mg Injection for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI. For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fracture:
Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognised risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10 – 40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Clostridium Difficile Diarrhea:
Published observational studies suggest that PPI therapy may be associated with an increased risk of *Clostridium difficile* associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency:
Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Interference with Laboratory Tests:
Increased Chromogranin A (CgA) level may interfere with investigations for neuroendocrine tumours. If the patient(s) are due to have a test on Chromogranin A level, Vaxcel Omeprazole 40mg Injection treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see section Pharmacodynamic). If CgA and gastrin levels have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

INTERACTION WITH OTHER MEDICAMENTS:

Effects of omeprazole on the pharmacokinetics of other drugs:
The following combination with Omeprazole powder and solvent for solution for injection should be avoided: ketoconazole and itraconazole. Omeprazole might influence the absorption of other drugs due to its effect on the gastric pH. The dissolution of ketoconazole tablets in the stomach is adversely affected in the pH of the gastric juice increases as a result of drug treatment (antacids, secretion-inhibiting agents, sucralfate). This leads to ineffective plasma concentrations of ketoconazole. During concomitant administration or omeprazole and itraconazole the plasma concentration and AUC of itraconazole are reduced by approximately 65%, probably as a result of poorer absorption, which is dependent on pH. Omeprazole inhibits the enzyme CYP2C19 and therefore increased plasma levels of the drugs maybe necessary. During concomitant administration of clarithromycin or erythromycin and omeprazole the plasma concentrations of omeprazole were increased. The plasma concentrations of omeprazole are not influenced during concomitant administration with amoxicillin or metronidazole.

Effects of other drugs on the pharmacokinetics of omeprazole:
Drugs inhibiting the enzymes CYP2C19 or CYP3A (HIV protease inhibitors, ketoconazole, itraconazole) might increase the plasma concentrations of omeprazole. No interactions between omeprazole and antacids, theophylline, caffeine, quinidine, lidocaine, propanolol, metoprolol or ethanol have been detected.

Incompatibilities:
No known incompatibility when recommended instruction are followed.

STORAGE:

Store below 30°C. Protect from light. Store in original box. The reconstituted solution for injection and infusion is stable up to 12 hours (concentration 4mg/ml in citrate buffer, water for injection, 5% w/v dextrose solution for injection) and 24 hours (concentration 0.4mg/mL in 0.9% w/v sodium chloride solution for injection and 5% w/v mannitol) at 25°C.

KEEP OUT OF REACH OF CHILDREN
JAUHI DARI KANAK-KANAK

PACK QUANTITIES:

Each combination pack contains 2 vials. 1 vial Vaxcel Omeprazole 40mg Injection. 1 vial Vaxcel Citrate Buffer 10ml as a solvent.

Further information can be obtained from pharmacist, physician or the manufacturer.

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